Capabilities Return: What Unlearning Leaves Behind, and What Warning Remains

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Abstract

Scope caveats first: every experiment here is at toy scale (two- to six-layer transformers on synthetic languages), the re-emergence arms use n=3 seeds extended to a pre-registered n=10 confirmation, one capability family (induction / in-context copying) carries the timing results, and one unlearning family (data-space active unlearning) is tested. Within that scope, every claim below was pre-registered with kill criteria before its data existed, scored by sealed one-shot scorers against committed artifacts, and survived (or is reported as killed).

In a companion paper we showed that a capability's *first* emergence can be forecast: a mechanistic precursor (the previous-token head) leads the induction capability by a config-invariant fraction of training, with certified false-alarm rates. Here we ask what happens *after* a capability is removed. Three findings. (1) *The precursor is a prerequisite, not the clock.* Supplying it at initialization causally accelerates emergence 1.75× (complete separation, p = 1.18e-09; placebo and near-miss priors do not accelerate), a *sub-threshold* seed captures most of the effect, and the dose–response is U-shaped: over-supplying a converged pattern is worse than seeding a weak one. (2) *Fingerprint monitoring of a guarded model gives certified detection with a characterized blind spot.* A library of capability fingerprints captured from an unguarded specimen detects faithful and relocated returns with zero false alarms on capability-blocked streams, but a return re-formed in an orthogonal representation defeats the frozen fingerprint entirely (behavior 0.948 vs. fingerprint readout 0.002). (3) *Unlearning removes the behavior, not the scaffold — so capabilities return.* After active unlearning drives copying to chance (and in-distribution copying *negative*), the precursor survives at 0.951–0.960; re-teaching restores the capability in 150 steps versus 6,300 for first emergence (42× faster). Denying the surviving scaffold doubles time-to-return (ratio 2.0; pre-registered confirmation at pooled n=10) — there is no free disguise — yet the denied return remains 21× faster than first learning, because everything else persists. The safety reading is uncomfortable and, we believe, decision-relevant: behaviorally unlearned is not removed; re-exposure recovers capability on timescales far shorter than first learning with or without the original circuit; and monitors of guarded models get certified *detection* but only fractional, not absolute, *warning*.

1 Introduction

A frontier lab that removes a dangerous capability from a model and then continues training it — fine-tuning, continual learning, tool-use adaptation — inherits a monitoring problem: *will the removed capability come back, and can we see it coming?* This paper studies that problem end-to-end at toy scale, where every quantity is measurable and every intervention is clean, as the third paper in a program on emergence timing [Howe, 2026a,b].

The companion result this builds on: for induction — the canonical in-context copying capability [Olsson et al., 2022] — a mechanistic precursor (the previous-token attention head) forms well before the behavior, leads it by a roughly config-invariant fraction of training, and supports calibrated, false-alarm-certified forecasts of *first* emergence. The natural extension to the guarded-model problem seems obvious: capture the capability's internal fingerprint while it exists, remove the capability, and watch for the fingerprint's return.

We report what actually survives when that idea is subjected to the certification discipline of the program (pre-registration with kill criteria, manufactured capability-blocked negatives, sealed one-shot scoring). The result is a reshaped picture:

Part I (causality). The precursor is causally load-bearing but is not the clock. Supplying it at initialization accelerates emergence 1.75$\times$ with complete seed separation; an equally aggressive task-useless prior and a near-miss prior (offset by one position) do nothing. A prior *weaker than our own detection threshold* captures most of the effect, and the dose–response is U-shaped with an interior optimum near the natural formation strength: the network amplifies a weak seed and *fights* an over-strong one.

Part II (monitoring). A fingerprint library captured from an unguarded specimen yields a monitor with certified false-alarm behavior — silent on omission and on an adversarial matched-data stream in which the capability's training signal is present but unlearnable — and it catches faithful returns and returns that relocate to a different attention head. Its blind spot is representational: a return forced to re-form orthogonally to the captured subspace restores full behavior while the frozen fingerprint reads noise.

Part III (unlearning and return). Active unlearning that drives the behavior to chance — and beyond, to *anti*-copying in distribution — leaves the precursor scaffold intact at 0.951–0.960 attention mass. Re-emergence is therefore not re-learning: it is re-composition on surviving infrastructure, 42$\times$ faster than first emergence. Denying the scaffold (which requires double the bias strength that suffices during formation — converged structure resists development-strength interventions) restores a relative cost: the denied return takes 2.0$\times$ as long. But it is a relative law only; in absolute terms the denied return is still 21$\times$ faster than first learning.

We also report what the discipline caught along the way: a guard that passed a naive reading but had only unlearned the capability *in distribution*; a "disguise" arm whose burn was silently defeated by a converged head; and an early loss-dynamics detector (in a companion rung) that passed every standard evaluation while reading the data mix rather than the capability. We believe the certification regime — not any single positive result — is the transferable contribution.

2 Setup and certification discipline

Substrate. Decoder-only transformers (2 layers, d=256, 4 heads, context 256 for the timing substrate; up to 6 layers for the multi-capability monitor) trained on synthetic Markov languages in which a fraction of sequences repeat their prefix at a variable offset, making match-based induction the only general solution. The behavioral event is an absolute crossing: copy advantage ≥ 2.0 nats (the difference in cross-entropy between first and second halves of a repeated random probe), sustained for two consecutive evaluations. First emergence on this substrate: median 6,300 steps (n=30).

Instruments. The precursor read is the maximum over heads of mean attention to the immediately preceding position (previous-token score); the capability's own circuit read is the prefix-matching score; the composed anchor — the only alarm rule in the program that has never false-alarmed anywhere it was scored — is the conjunction of the precursor and an in-distribution behavioral ramp in the same evaluation.

Discipline. Every rung is pre-registered (predictions and kill criteria committed before the data exists), scored by a sealed one-shot scorer that refuses to run on partial fleets or twice, and adjudicated against manufactured capability-blocked negatives: data ablation (no repeats), architectural blocking (one layer), and adversarial matched-data streams (the capability's marker present at identical statistics but with scrambled targets, so a detector that fires is reading the data condition, not the capability). Manipulation checks are scored before outcomes and can void an arm; two arms in this paper were voided and re-run this way.

3 Part I: the precursor is a prerequisite, not the clock

Causal seeding. We add a fixed additive attention-score bias $B[i, i-1]=\beta$ to a single layer-0 head at initialization — an architectural supply of the previous-token pattern — and measure emergence timing against 30 controls. At β=8: median 3,600 versus 6,300 (43% acceleration, 1.75×), with *complete separation* (every scaffolded run beat every control; exact one-sided permutation $p = 1.18\text{e-}09$). A task-useless placebo of identical magnitude (bias to position 0) lands at 6,725 and a near-miss prior ($i-2$) at 6,725 — neither accelerates; both are, if anything, slightly slow, consistent with occupying the head that would otherwise become the precursor. A no-repetition corner (scaffold supplied, training signal ablated) produces zero events: architecture cannot manufacture the capability without data.

The pre-registered clock hypothesis failed. If the precursor were the sole rate-limiter, supplying it should compress emergence to the post-anchor residual (∼1,000 steps). Observed: 3,600 — the bet missed by 3.7×. The diagnosis (post-hoc, then confirmed prospectively in the dose rung's floor bet, which *fired its own kill*): emergence is gated by more than one prerequisite, and deleting the slowest merely exposes the next. We report this as the central interpretive lesson of Part I: *interventions on named precursors re-time emergence but do not schedule it.*

Dose–response is U-shaped; a sub-threshold seed suffices. Sweeping $\beta \in \{1, 2, 3, 4, 6, 8\}$ (5 seeds each): medians 3,825, 3,375, 3,350, 3,425, 3,525, 3,600. Two facts matter. First, β=1 imposes an initial previous-token weight of 0.079 — *below* the 0.10 threshold our own campaign uses to declare the precursor present — and still accelerates emergence to 3,825: the network amplifies the sub-threshold hint past threshold within 100 steps. "Supplying the precursor" therefore does not require supplying a converged head; a seed the network grows itself is enough, and "precursor present" is better understood as a rate than a state. Second, the curve has an interior optimum near β=3 and *rises* toward β=8: the ascending arm is seed-paired with zero reversals (15/15). Per-head instrumentation shows a sharp adopt-versus-fight crossover: at $\beta \leq 3$ the network amplifies the supplied head above its imposed value in every run; at $\beta \geq 4$ it suppresses it in every run. Too-strong priors are actively resisted.

Placement. The identical bias on layer 1 — where a previous-token head cannot feed the induction circuit — lands at 6,700 (no acceleration), and the biased head never becomes an induction head. We register the honest limit: this is consistent with the composition story but under-determined (the arm is statistically indistinguishable from the placebos, so "layer-1 is a bad site for any bias" is not excluded; the discriminating layer-1 placebo is future work).

4 Part II: fingerprint monitoring — certified detection, characterized blind spot

We instantiate the guarded-model monitor end-to-end on a two-mechanism library (induction, with an attention-alignment fingerprint; bracket-depth counting, with a rank-4 interior-residual subspace fingerprint captured by one-shot ridge fit from the frozen specimen at decode R^2=1.0). Protocol:

train the unguarded specimen; capture the library from frozen weights; guard each skill by active unlearning (answers scrambled at matched marker statistics) until behavior and structure sit at chance; then continually train under watch streams while reading each fingerprint's proximity inline on fixed probe banks — never on the training stream.

Certified detection. The composed alarm (structural proximity above an absolute band set from capability-blocked baselines, AND a behavioral ramp) is silent on every negative stream — omission and marker-scramble, both skills — and fires on faithful returns. **Relocation is caught.** Burning the fingerprint's home head forces the returning skill onto a different head; the frozen-home read misses (0.335) while the max-over-heads read catches it (0.661). **Orthogonal re-formation is not.** Re-teaching the counting skill while continuously erasing the captured subspace restores full behavior (0.948) while the frozen fingerprint reads 0.002 and the alarm never fires — with the negatives silent, this is a verified blind spot, not a blind monitor. A capability can return wearing a representational disguise the specimen fingerprint cannot see; broad reads defeat cheap disguises, not this one.

No early warning for re-emergence at this depth. In the pilot's substrate the guarded skills returned within a few hundred steps and alarms were coincident (lead 0) or lagging (-1400 for the subspace read) — which motivated Part III's move to the slow substrate where first-emergence timing is understood.

5 Part III: unlearning removes the behavior, not the scaffold

Unlearning is distribution-local (a caught manipulation, kept as a finding). Our first guard — training on sequences whose repeats *lie* (copied spans permuted) — drove in-distribution copying strongly negative while out-of-distribution copy advantage plateaued at $\sim$1.1–1.4 nats: the model learned "repeats lie *here*" while generic match-and-copy survived. The frozen validity bar rejected the guard; guard v2 adds lying repeats *in the surviving circuit's own distribution* and passes on all seeds (copy advantage ≤ 0.09 nats, in-distribution negative).

The scaffold survives complete behavioral removal. After a valid guard, the previous-token scaffold reads 0.951–0.960 — essentially the specimen value — on every seed. Unlearning deleted the capability's output behavior and left its upstream prerequisite untouched.

Hence re-emergence is re-composition, not re-learning. Re-teaching restores the capability in a median of 150 steps versus 6,300 for first emergence: 42× faster. The composed anchor leads by 75 steps — consistent across every run, but three evaluation ticks in absolute terms. The firewall holds (zero alarms on guard-hold streams).

Converged structure resists development-strength interventions. Our first "denied" arm burned the surviving scaffold head with the same β=8 bias that reliably burns *forming* heads; the converged head's learned margin defeated it (residual previous-token mass 0.57–0.70 at step 0), and the arm was voided by its manipulation check. β=16 silences it (step-0 mass ≤ 0.038). The asymmetry — β=8 suffices during formation, β=16 is needed at convergence — is the deployment-relevant mirror of Part I's adopt/fight crossover.

The compensating law: no free disguise, but a cheap one. With the scaffold genuinely denied, the return must regrow a precursor elsewhere, and pays for it: median 300 steps versus 150 faithful — ratio 2.0 at n=3. A disguised return costs real relative time — there is no free disguise — but the absolute refuge is thin: even scaffold-denied, the return is 21× faster than first learning, because the language, the downstream circuit, and everything else the capability composes with persist through unlearning.

6 Related work

Activation-clamping during training established qualitatively that supplying or removing induction subcircuits moves the phase change [Singh et al., 2024]; their intervention is activation-level surgery across seeds, without placebo or near-miss controls, dose response, or timing-law tests. Data-side steering of induction-circuit formation timing [pat, 2026] is the data-dual of our architectural lever. Architectural priors that hard-wire local-shift structure — relative-position biases [Press et al., 2022], short-convolution and token-shift primitives [Fu et al., 2023] — are the deployment ancestors of our β-bias; none measures capability arrival time. On the monitoring side, our fingerprints are standard instruments (attention pattern scores [Olsson et al., 2022]; probe subspaces); the contribution is the certification regime around them. The unlearning findings connect to the growing literature on shallow unlearning and rapid relearning [Lynch et al., 2024]; our contribution is mechanistic: *what* survives (the precursor scaffold, at full strength), *why* return is fast (re-composition on surviving infrastructure, quantified against first-emergence timing on the same substrate), and what denial costs (a relative factor of $\sim$2.0, not an absolute refuge).

7 Limitations

Toy scale throughout; one capability family carries the timing claims; one unlearning family (data-space active unlearning) is tested — weight-space methods may leave different residues; re-emergence arms are n=3 extended to a pre-registered pooled n=10 confirmation with the decision line frozen at the original kill threshold; the compensating-law ratio sits at its pre-registered bar and is reported as boundary-grade evidence; the 25-step evaluation grid limits timing resolution; the layer-1 placement reading is under-determined; anchor lead fractions are not conserved across arms and we make no law claim for re-emergence warning time.

8 The kill ledger

Claims that died, pre-registered bets that failed, and manipulations the discipline caught — reported with the same prominence as the positives: the clock hypothesis (missed by 3.7×); the two-gate fixed-assembly model (killed by its own prospective floor bet in the dose rung); "placebos are inert" (they delay); the seed-stickiness arm (unreachable bar — arithmetic, not evidence); the loss-dynamics detector that passed every standard evaluation while reading the data mix (killed by a matched-data capability-blocked cell); guard v1 (distribution-local unlearning); the β=8 burn (defeated by a converged head); and the fingerprint early-warning claim for re-emergence (no absolute warning exists at this scale).

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